PIANOID: 3D PIANIST BODY MOTION GENERATION FROM MIDI

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ABSTRACT

Pianist performance is a full-body activity, with torso sway, shoulder lift, and head tilt conveying expression alongside the notes. Yet computational models of pianist body motion have so far addressed only hands or 2D upper bodies. We present **Pianoid** [1], the first 3D 26-joint pianist body motion system from MIDI. Pianoid is a four-stage hybrid pipeline whose body partition and diffusion placement are derived from a coupling analysis of the MOSA performance corpus. MPJPE (Mean Per Joint Position Error) rewards the near-static collapse of mean-regressed baselines. To surface this failure mode we report per-axis expressiveness ratios aggregated as Δ_1, validated by a dual perceptual study (non-expert N=28, expert N=9). Non-expert viewers prefer Pianoid over every baseline and ablation, while still preferring the ground truth. Experts attribute the residual gap to fine timing and articulation rather than under-expression. Pianoid establishes note-level pianist body motion as a tractable MIR task via data-derived hierarchical modeling with selective diffusion.

1. INTRODUCTION

When you watch a pianist perform, the music you hear is inseparable from what you see. The torso sways on a climactic chord, the shoulders lift before a leap, and small postural resets punctuate the spaces between phrases. Visual expression is critical for conveying musical ideas to the audience [1]. Prior work has shown that expressive pianists increase torso sway and head rotation in synchronization with musical phrasing [2], and that audiences perceive performances as more expressive when body motion is visible.

Yet computational models of pianist body motion have remained underexplored. Data-driven work has focused on hand/finger motion [3–5], or used limited body models (e.g., 2D, 8 upper-body joints) [6]. Other systems condition on audio rather than symbolic music [7]. No prior system has generated *3D full-body* pianist motion from MIDI.

To address this gap, we present Pianoid, the first system that generates 3D 26-joint full-body pianist motion from MIDI. Our input is the pitch and note-on/off events from MOSA (Music mOtion with Semantic Annotation) [8], without per-note velocity (dynamics). Pianoid is a four-stage hybrid pipeline. Stage 1 regresses wrists, Stage 2 generates torso sway, Stage 3 regresses elbows, and Stage 4 regresses the lower body. Stochastic diffusion is reserved for Stage 2 (the single one-to-many sub-problem). The other three stages use deterministic regression. The decomposition and per-stage generators are derived from a computational analysis of music-body coupling on 272 MOSA sequences (all 8 pianists), rather than fixed *a priori*.

That coupling analysis revealed two findings that shape the design. *First*, shoulders couple more tightly to the torso than to the ipsilateral wrist, so we move the shoulder out of the arm chain and into the backbone. This redraws a body-partition convention that prior hierarchical motion models [9, 10] adopt to keep the arm (shoulder through wrist) as a single "arms" unit. *Second*, deterministic regression collapses to the conditional mean on expressive one-to-many targets such as torso sway, a failure that MPJPE under-penalizes and can even reward. We localize it to Stage 2 and fix it by reserving diffusion for that single sub-problem. To expose what MPJPE hides we report per-axis *expressiveness ratios* (range, speed, jerk), aggregated as a symmetric log deviation Δ_1 that penalizes over- and under-expression equally. A perceptual study with both non-expert viewers (N=28, 2AFC) and pianist experts (N=9, thematic interview) complements the quantitative results.

Our contributions are fourfold. First, we deliver the first system to generate 3D 26-joint full-body pianist motion from MIDI alone. Second, we derive a body partition from data that inverts the conventional "arms" boundary (F3). Third, we place stochastic diffusion selectively, justified by per-stage one-to-many analysis (K1). Fourth, we introduce per-axis expressiveness ratios aggregated as Δ_1, validated by a dual non-expert/expert perceptual study in which the metric tracks expert judgment where MPJPE inverts the ranking.

2. RELATED WORK

Prior piano motion generation has remained at hand granularity, splitting into two paradigms. *Kinematic-based* models predict joint trajectories from mocap data, ranging from Li et al.'s [6] CNN-LSTM for 2D upper-body skeletons

[1] Project page with code, pretrained checkpoints, analysis scripts, expert-study prompts, and an audio-synced interactive comparison of the 12 non-expert 2AFC clips against ground truth and all baselines, https://joonhyungbae.github.io/pianoid/.

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(ISMIR 2018, the closest prior work in body-level scope) to recent large-scale hand diffusion models including PianoMotion10M [3], FürElise [4], and S2C [5], the last introducing dual-stream diffusion for bimanual coordination. *Physics-based* RL on simulated pianos trains policies for accurate key contact, including RoboPianist [11] and follow-ups [12, 13], but produces robot-like motion that prioritizes task success over expressiveness. Audio-driven body synthesis exists [7] but does not condition on symbolic music. Li et al.'s 2D/8-joint subjective evaluation reported no significant difference from real motion on 75% of pieces, suggesting low-dimensional pianist motion approaches subjective ceiling. Pianoid targets the harder 3D/26-joint setting where this ceiling is not yet attained, beyond the hand-only, 2D/8-joint, and audio-conditioned scopes of prior work.

Music-conditioned full-body motion has been explored in adjacent settings. Dance-to-motion methods [14–16] generate free-form full-body movement conditioned on audio. They lack piano-specific biomechanical constraints (seated pose, keyboard reach) and target aesthetic choreography rather than note-level control. Motion diffusion models [17] provide the core generative machinery we adopt for Stage 2. Stochastic sampling from a learned distribution is the standard remedy when deterministic regression collapses to the conditional mean on one-to-many targets. Recent extensions (MoFusion [18], MotionGPT [19], PriorMDM [20], GMD [21], and OmniControl [22]) improve controllability through learned representations (tokenization, latent priors) or spatial guidance signals, but remain monolithic. None decomposes the target body hierarchically according to data-driven coupling structure. None addresses cross-stage distribution shift either, since no cascade is exposed. Our Single-Stage MDM+zs ablation (Section 5) isolates what the decomposition adds on top of this machinery, since it retains every Stage 2 component in one monolithic model and still under-expresses on every axis. Hierarchical body decomposition has been explored in text-to-motion, notably by ParCo [9] and Ghosh et al. [10]. Both partition the body so that arms (shoulder+elbow+wrist) form a unit separate from the torso. ParCo explicitly credits this partition to the SMPL/MMM kinematic trees rather than to data. Our partition is instead *derived from empirical coupling statistics* that place shoulders with the torso rather than with the arms (F3). A fixed taxonomy would miss this regrouping. Across these strands, no prior system applies a data-driven hierarchical decomposition to 3D full-body pianist motion from MIDI. Pianoid addresses this combination.

Performance science has long linked visual motion to musical expression. Audiences judge expressiveness from body motion [1], "ancillary gestures" convey intention beyond sound production [23], and pianists' movements encode motor planning and anticipation [24], expressive intention and structure [25], phrase boundaries [26], and expressive timing [27]. Experienced pianists also show greater torso and head motion when playing expressively [2]. These studies are largely descriptive or limited in scale. Section 3 extends them computationally across all 8 MOSA pianists, with each finding mapped to a specific pipeline decision in Pianoid.

3. DESIGN RATIONALE: MUSIC-BODY COUPLING

We analyze 272 sequences from MOSA (8 pianists, 60 fps, 26 joints, Henle Verlag difficulty grades 3–8) to characterize how symbolic music drives the pianist's body. Several phenomena below have prior qualitative documentation in performance science [24, 28].

Correlation protocol. All coupling statistics are Pearson correlations on raw per-axis joint-coordinate series, without filtering or detrending. For F3 we pool a piece's performer trajectories along time and average $|r|$ over the 24 shoulder–torso pairs (2 shoulders $\times$ 4 torso markers $\times$ 3 axes) and the 6 ipsilateral shoulder–wrist pairs (2 sides $\times$ 3 axes), taking absolute values because the sign of a coordinate correlation tracks marker laterality rather than coupling strength. Cascade statistics (F2, F6) use the lateral axis, where torso sway lives. To separate biomechanical coupling from shared musical drive, wrist→C7 is recomputed after linearly residualizing both series on the per-frame MIDI pitch centroid. F4 and F5 are Benjamini–Hochberg corrected, all numbers use the full 272 sequences, and every F1–F6 effect replicates in sign on train-only performers (yp01–yp05), ruling out test-set leakage.

Table 1 lists the raw inter-joint correlations, and Table 2 consolidates the six analysis findings (F1–F6) and three additional kinematic observations (K1–K3) with their pipeline implications. The decisive finding (F3) is that shoulders couple more tightly to the torso (piece-mean $|r|$ 0.65–0.87, mean 0.81) than to the ipsilateral wrist (0.25–0.48, mean 0.37). Per-piece direction (24 shoulder–torso vs 6 shoulder–wrist axis pairs each, two-sided binomial sign test) favors torso on $14/14$ pieces (p=1.2e-4). This inverts the SMPL/MMM-inherited "arms" partition that prior hierarchical motion synthesizers [9, 10] adopt, which keeps shoulder+elbow+wrist as a single unit. We move shoulders into the torso group, aligned with Bernstein's coordinative-structures hypothesis [29], and let it drive our Stage 2 backbone.

The other findings group into three themes. In the biomechanical cascade (F2, F6), wrist-driven torso lean survives pitch partialling (r=0.65) and spine lateral anticipates keystroke by $\sim$65 ms [24], prescribing the hands→body direction.

Hierarchical divergence (F1, F4, F5) appears as a $6\times$ motion-variance range across parts and as opposite density responses (wrist $+17\%$ vs. head -16%, consistent with Furuya's trunk-stabilization principle [28]). The parts also require distinct onset-density windows. The wrist–head sign reversal is the critical signal. A single conditioning pathway cannot match opposite responses to the same musical feature. Even a modest divergence (Cohen's $d\approx$0.29) suffices to motivate separate streams.

Target-distribution shape (K1–K3) varies. Torso sway

Table 1. Inter-body-part Pearson r, mean over 272 sequences. Slashes denote variants (L/R hands for wrist–lateral, raw vs MIDI-partialled |MIDI for wrist–C7). Bold rows show per-piece $|r|$ ranges across the 14 pieces.

Signal pair	Type	r
MIDI pitch – Wrist lateral	Musical	0.37/0.44
MIDI pitch – Spine C7	Musical	0.100
Wrist – C7 (raw / |MIDI)	Biomech.	0.658 / 0.652
Head – C7 / Spine – Pelvis	Rigid/Post.	0.941 / 0.861
Wrist – Elbow	Geometric	0.946
Shoulder – Torso	**Anatomical**	**0.65–0.87**
Shoulder – Wrist	Functional	0.25–0.48

Table 2. Analysis findings (F1–F6) and kinematic observations (K1–K3, below dashed line), each with key statistic and pipeline implication.

	Finding	Key statistic	⇒ Design
F1	Kinematic hierarchy	wrist 45.9 / pelvis 7.1 mm std (6×)	4 stages
F2	Biomech cascade	wrist→C7 r=0.66 (|MIDI 0.65; MIDI→C7 0.10)	S1→S2→S3/S4
F3	**Anat. dominates func.**	**shldr-torso** 0.65–0.87 **vs wrist** 0.25–0.48 (14/14, p=1.2**e-**4)	**S2 w/ backbone**
F4	Wrist/body divergence	density↑: wrist +17%, head −16% (d≈0.29)	separate W/B
F5	Per-part temporal scales	wrist 0.5s / backbone 1.0s, staccato d=0.38	per-stage od
F6	Anticipation/keystroke	spine leads 65 ms, onset Z-desc. 50.6%/47.9% (χ^2=656)	S1 Z log-ratio
K1	Torso sway 1-to-many	performer variance 9.1×	S2 diffusion
K2	Elbow swivel unimodal	±12°	S3 deterministic
K3	Lower body passive	spine→pelvis r=0.86	S4 FiLM on spine

is one-to-many, elbow swivel is unimodal, and the lower body passively follows the spine. Together these dictate the four-stage split, diffusion placement, and cascade direction.

4. MODEL

The Pianoid model adopts a four-stage hybrid pipeline (Figure 1). Stochastic diffusion is used only for the one-to-many sub-problem (Stage 2 backbone). Deterministic regression handles the constrained or unimodal stages (Stages 1, 3, 4). Stage boundaries, generator choice, and representations are each justified by a specific finding from Section 3, with the corresponding ablation reported in Section 5.3. The stage *order* is likewise read off the analysis rather than chosen. Wrists predict C7 at r=0.66 while MIDI predicts it at 0.10 (F2), so the hands must lead the body, and the spine drives the pelvis passively at r=0.86 (K3), which places the lower body last.

Stage 1 maps MIDI to wrists with a BiLSTM. Concatenated MIDI features (piano roll, onset mask, per-key onset density, duration/IOI ratio, 352-dim) are mapped to wrist positions (6-dim) via a BiLSTM (hidden 512, 6 layers). Loss combines L1 position and velocity with scale-invariant *log-ratio* losses on Z-axis range and std, which preserve the vertical keystroke signature identified in Section 3 (50.6% wrist descent at onsets, F6). MPJPE-only minimization collapses wrist-Z range. The Z log-ratio losses recover it at a 10 mm MPJPE cost (Table 5 C). This trade-off recurs at Stage 2 with diffusion.

Stage 2 maps predicted wrists and onset density to a *10-joint backbone* (head 4 + torso 4 + shoulders 2, 30-dim total) using a diffusion model (MDM) with per-joint-per-axis z-score target normalization and DDIM sampling at inference. Two architectural choices, both empirical.

(i) Grouping shoulders with the backbone (not with arms) is motivated by the shoulder-torso correlation 0.65–0.87 versus shoulder-wrist 0.25–0.48 (piece-mean $|r|$, Section 3). Placing shoulders in the arm stage (the functional grouping that prior work assumed) creates a silent gradient cancellation between shoulder placement and arm-loss gradients that we diagnose in Table 5 B.

(ii) Diffusion rather than regression. Torso sway is a one-to-many mapping. Two performers given the same MIDI and wrist trajectory can produce very different sway trajectories (performer-level variance is 9.1× within the dataset). A deterministic BiLSTM trained on this mapping collapses to the conditional mean. In our experiments, this means ∼34% of GT speed and range ratio ∼0.48, i.e., a near-static upper body. MDM [17] samples from the learned distribution. At 45 DDIM steps, it restores expressiveness to ER-range 1.12, ER-speed 0.92, and ER-jerk 0.34. We accept a ∼20 mm MPJPE increase as a deliberate trade-off (Table 5 A). MPJPE-minimizing models suppress expressive motion.

Two implementation choices complete the design. *Per-axis z-score target normalization with scale-invariant log-ratio range/std losses* balances learning across axes, since in raw coordinates the high-variance lateral axis dominates and the vertical lift axis under-trains. *DDIM step count* then acts as an expressiveness dial, with fewer steps under-expressing and more over-expressing. Sweeping 10–50 steps on the held-out validation split (34 sequences) leaves Δ_1 flat at 30/45/50 (within 0.03), and a pre-specified $|\ln$ ER-range$|$ tiebreaker selects 45. All test-set conclusions are robust to defaulting to 30 (MPJPE −1.4 mm, Δ_1 −0.02). The test split (80 sequences) is used only for final reporting.

Stage 3 maps the predicted wrists, chest, and shoulders (18-dim spatial), plus onset density and pitch range, to elbow positions (*relative to the predicted ipsilateral shoulder*) via a BiLSTM. Deterministic regression dominates diffusion here. An MDM under matched architecture and conditioning at the same training compute (Table 5 B last row) reaches 293.9 mm MPJPE with 12.25× over-expressed jerk. The gap is stable across a DDIM-25/50/100/200 inference sweep (199–306 mm). The elbow's swivel angle is unimodal (±12°) once wrist and shoulder are fixed, so diffusion's stochastic sampling becomes a liability rather than an asset.

Two fixes distinguish this stage from a naive implementation. *Training against predicted rather than ground-*

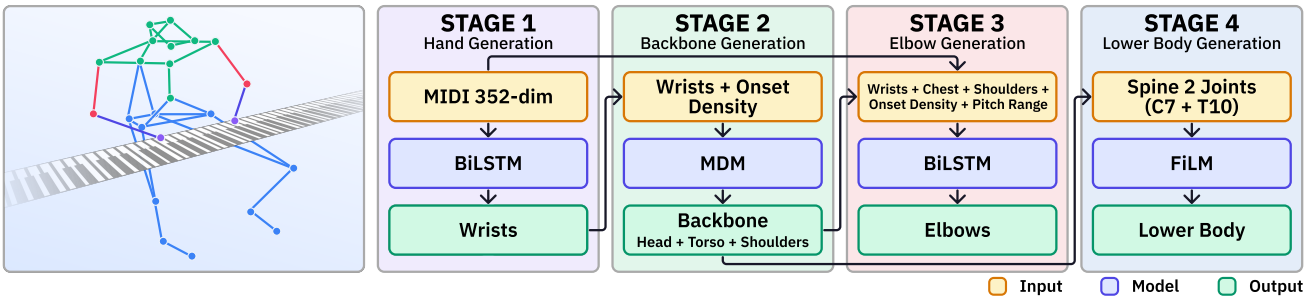


Figure 1. **Pianoid pipeline (4-stage hybrid).** Diffusion is reserved for the one-to-many backbone (Stage 2), with deterministic regression at Stages 1, 3, and 4. The shoulder grouping at Stage 2 follows F3 (Table 1).

truth shoulders mitigates the cascade analog of the teacher-forcing gap (cf. scheduled sampling [30] and frozen-upstream training in PianoMotion10M [3]). On our stack the gap is small (Table 5 B, 3 seeds), improving anatomical Stage-3 MPJPE by 3.3 mm (86.2→82.9), neutral under functional, with L/R asymmetry uniformly < 2 mm, and we retain it as defensive practice for cascades with less-aligned upstream stages. *Shoulder-relative coordinates* keep shoulders as conditioning input but not as a target, since Stage 2 already produces them. This avoids a silent gradient cancellation in which absolute-shoulder and relative-elbow losses interfere through the shared shoulder term. Light log-ratio range/std losses (weight 0.2) on the absolute elbow then restore GT-matching temporal dynamics (elbow-only ER-jerk 1.02).

Stage 4 maps the predicted spine to the lower body via FiLM modulation, taking C7+T10 (6-dim) input to 10 lower-body joints. Onset density and ARIA [31] features do *not* help here (+0.4 and +7.0 mm respectively, Table 5 D). This is consistent with the analysis finding that the seated lower body is nearly static and follows the spine passively rather than responding to musical features. All four stages are evaluated jointly in Section 5, with per-stage ablations isolating each design choice.

5. EXPERIMENTS

Our evaluation has three layers. MPJPE provides comparability with prior work. Per-axis expressiveness ratios aggregated as Δ_1 surface the mean-collapse failure mode anticipated by Section 3 (F4 wrist/head divergence, K1 torso 1-to-many), and serve as a sanity check rather than a stand-alone acceptance criterion. The dual perceptual study with non-expert viewers (N=28, 2AFC) and pianist experts (N=9, thematic interview) is the final arbiter.

5.1 Setup

MOSA provides 26-joint mocap at 60 fps over 8 pianists (yp01–yp08), 14 pieces (Henle 3–8), totaling 272 sequences split 158/34/80 train/val/test. The split is performer-disjoint (train yp01–yp05, val yp06, test yp07–yp08), except that for the two hardest pieces (*be1*, *tc1*) yp04/yp05 are routed to test (12/80 samples) to measure seen-performer unseen-piece generalization, with the remaining 68/80 measuring unseen-performer generalization. MIDI is reconstructed from MOSA note annotations (no per-note velocity, no finger articulation).

Table 3. Stage- and baseline-specific hyperparameters. All use AdamW (lr 1e−4), batch 16, 500 epochs, patience 50, val-MPJPE early stopping. *Li2018 uses weight-decay 10^{-8}. †EDGE uses window 480 frames.

Pianoid stage / baseline	hidden	layers	DDIM
S1 wrist BiLSTM (ours)	512	6	—
S2 backbone MDM (ours)	256	8	45
S3 elbow BiLSTM (ours)	256	6	—
S4 lower FiLM (ours)	128	4	—
Li2018 CNN-LSTM*	256	2	—
MDM-adapted	512	8	25
EDGE-adapted†	256	4	25
Single-Stage MDM+zs (abl.)	256	8	45
BiLSTM-on-backbone (abl.)	512	6	—

We evaluate against three *published* baselines (Table 3), each re-trained on Pianoid's training split with its released architecture and original loss and only the conditioning adapted to MIDI. Li2018 [6] is a CNN-LSTM adapted from 2D/8-joint to 3D/26-joint, MDM-adapted [17] is the motion diffusion model with MIDI conditioning, and EDGE-adapted [14] is the dance diffusion model with MIDI replacing Jukebox. Two *ablation* variants isolate the hierarchy and diffusion contributions. Single-Stage MDM+zs is monolithic but carries all Stage 2 components (per-axis z-score, log-ratio losses, DDIM 45), and BiLSTM-on-backbone is a deterministic regressor with the same I/O as Stage 2.

Beyond MPJPE, we report metrics that surface the failure modes we diagnose in Section 4. We introduce three *per-axis expressiveness ratios* (ER) comparing prediction $\hat{x}^{(a)}_{j,t}$ to GT $x^{(a)}_{j,t}$ along each axis $a \in \{x, y, z\}$, joint-axis-averaged. ER-range is the ratio of per-axis max−min, ER-speed the ratio of mean $|\dot{x}^{(a)}|$ (first finite difference at 60 fps), and ER-jerk the ratio of mean $|\dddot{x}^{(a)}|$ (third finite difference at 60 fps). All ratios equal 1 for GT-matched motion, with values below 1 indicating under-expression and above 1 over-expression, averaged over the 80 test sequences. Two piano-specific ratios complement these. *WZR* (wrist Z-axis range ratio, → 1) is a keystroke preservation measure, and *AccR* (backbone acceleration ratio, → 1) is the second-derivative analog of ER-speed.

We aggregate the three ER ratios by their symmetric

Table 4. Benchmark on 80 test sequences. MPJPE over 26 joints, ER over 10 backbone joints, all ratios targeting 1. **Bold**=best, underline=2nd. Ablations in Table 5.

Method	MPJPE↓	ER-rng	ER-spd	ER-jrk	WZR	AccR	Δ_1↓
Li2018*	**76.7**	0.66	2.92	13.1	0.17	10.5	1.36
MDM-ad.	88.2	0.82	0.46	0.25	0.38	0.25	0.79
EDGE-ad.	80.2	0.71	0.51	**0.69**	0.27	**0.54**	0.46
Pianoid	92.1	**1.12**	**0.92**	0.34	**0.76**	0.44	**0.42**
GT	—	1.00	1.00	1.00	1.00	1.00	0.00

*Li2018's 13×GT jerk follows from extending its 2D/8-joint architecture to 3D/26-joint output, not a claim against the original 2D/8-joint setting. Bootstrap 95% CIs for Pianoid (N=80, 10,000 resamples) are MPJPE $[88.8, 96.6]$, ER-rng $[1.07, 1.16]$, ER-spd $[0.86, 1.00]$, ER-jrk $[0.31, 0.36]$, WZR $[0.71, 0.82]$, AccR $[0.41, 0.47]$, Δ_1 $[0.39, 0.46]$. Baseline per-sample outputs were not retained, so these are descriptive intervals on sampling variability, and significance vs. baselines comes from the perceptual study (Section 5.4).

log deviation, $\Delta_1 = \text{mean}|\ln \max(\text{ER}, 10^{-6})|$ (natural log throughout). This formulation ensures over- and under-expression by the same factor (e.g. ER=0.5 and ER=2.0) incur equal penalty. The 10^{-6} floor guards against degenerate near-zero motion, and no aggregate ER triggered the clamp on our test set. Δ_1 is intentionally outlier-sensitive. A single axis with ER$\gg$1 inflates it (the elbow-MDM ablation row with ER-jerk 12.25× contributes $|\ln|$=2.5), so per-axis $|\ln \text{ER}|$ values are reported alongside in Tables 4 and 5. The three $|\ln \text{ER}|$ axes are weakly correlated across test sequences (max Pearson $|r|$=0.26), so Δ_1 aggregates near-independent dimensions.

Δ_1 is an aggregate sanity check. The 2AFC study (Section 5.4) is the principal statistical test, with per-participant paired Wilcoxon at Holm-adj. $p<10^{-10}$ vs. all baselines except EDGE ($p<0.01$). Baselines are kept monolithic (not injected into our 4-stage architecture). ER metrics are computed at the sequence level on each baseline's test-split inference outputs (Table 3 training). Baseline per-sample outputs were not retained from inference, so the bootstrap CIs in Table 4 report Pianoid's sampling variability descriptively rather than as substitute paired tests.

5.2 Benchmark Results

MPJPE rewards mean-collapse (Table 5 A). Both Δ_1 and the perceptual study (Section 5.4) confirm Pianoid's higher MPJPE is principled rather than a regression. Li2018's best-MPJPE (76.7 mm) is a near-static collapse (wrist-Z 0.17×GT, jerk 13×GT), mirroring our Stage 2 regression ablation (Table 5 A). Li2018 mean-collapses to win MPJPE, and MDM-adapted under-expresses on every ER axis despite stochastic sampling. EDGE is the closest baseline to GT on ER-jerk and AccR, but at 0.69 and 0.54 it still under-expresses there and more so on range and speed. All three baselines have $\Delta_1 \geq 0.46$. Pianoid wins outright on 4 of the 6 expressiveness metrics (ER-range, ER-speed, WZR, Δ_1). It is runner-up on the other 2 (ER-jerk and AccR, the two axes where EDGE is closest to GT), reaching Δ_1=0.42. Bootstrap CIs do not overlap any baseline on MPJPE, ER-rng, ER-spd, or WZR, while the EDGE Δ_1 CI touches ours at the boundary. Per-participant 2AFC Wilcoxon (Section 5.4) gives Holm-adj $p<0.01$ vs. EDGE and $p<10^{-10}$ vs. others.

A *Single-Stage MDM+zs* variant (Table 5 A) applies all Stage 2 components monolithically. It lowers MPJPE to 81.1 mm yet under-expresses on every ER axis (Δ_1=0.71 vs. 0.42). This confirms that hierarchy supplies expressiveness gains representation alone cannot.

Per-body-part MPJPE tracks the Section 3 coupling hierarchy that the four-stage decomposition targets. Constrained joints (pelvis 47.4, wrists 64.1 mm via S4/S1) are far more predictable than expressively-free ones (head 111.1 mm via S2), with mid-tier groups between (torso 79.1, elbows 82.9, shoulders 88.0, legs 88.5 mm). Head exceeds shoulder because head std (33.6 vs. 23.2 mm) widens the diffusion posterior, and a deterministic head would lower MPJPE but produce trivial motion (Stage 2 ablation A).

5.3 Ablation: Analysis-Driven Fixes

Table 5 presents per-stage ablations (blocks A–D map to Section 3 findings K1, F3, F6, K3). Each surfaces a failure mode MPJPE masks and per-axis ER reveals, confirming MPJPE alone is insufficient. The MPJPE column is on all 26 joints in block A, on elbows in B, on wrists in C, with pelvis and legs given separately in D.

(A) Stage 2 backbone (regression vs. diffusion). A deterministic BiLSTM backbone wins MPJPE by ∼20 mm but is near-static (Δ_1=1.41). MDM+zs at 45 DDIM restores expressiveness (Δ_1=0.42). DDIM step count is a continuous expression dial, plateauing at 0.40/0.42/0.43 for 30/45/50 steps. Selection of 45 was made on the val split per the pre-specified tiebreaker $|\ln$ ER-range$|$ (Section 4). Defaulting to 30 leaves every conclusion intact.

(B) Stage 3 elbow. We ran the 2×2 over 3 seeds (per-cell std $<$ 1.5 mm). Anatomical regrouping is the dominant factor, raising elbow Z-range 10× and ER-jerk 4× by removing a silent $r=-0.975$ shoulder-Z / relative-elbow-Z anti-correlation. Predicted-input training improves anatomical MPJPE by 3.3 mm (86.2→82.9), neutral on functional, with L/R asym $<$ 2 mm. The fifth row (MDM at matched compute, +211 mm) confirms diffusion is the wrong inductive bias for unimodal elbow swivel.

(C) Stage 1 wrist loss (same MPJPE/expressiveness trade-off). Vanilla L1+velocity (v1) yields best MPJPE but collapses wrist-Z range. Speed matching (v2) worsens it, as lateral axes dominate. Only Z-axis log-ratio losses (v3) recover ZRngR 0.85 at a 10 mm MPJPE cost. MPJPE is insufficient at *every* stage.

(D) Stage 4 lower body (simpler wins for passive joints). FiLM on predicted-spine alone reaches 70.7 mm. Adding onset density or ARIA [31] *hurts*, and swapping FiLM for BiLSTM costs 2.0 mm. The seated lower body follows the spine passively, so music features add noise.

5.4 Perceptual Evaluation

A web-based 2AFC study ranks GT, Pianoid, and baselines as 26-joint skeletons over a keyboard plane, synced to audio. Stimuli are 12 clips per pair-type (9 main with

Table 5. Per-stage ablation. **Bold**=chosen. Block B is the 2×2 over Stage 2 grouping and Stage 3 training scheme, plus an MDM variant.

Variant	MPJPE↓	Block-specific cols		
(A) Stage 2 backbone. Cols: ER-rng, ER-spd, ER-jerk.				
Monolithic MDM+zs	81.1	0.67	0.50	0.35
BiLSTM regression	**71.8**	0.48	0.34	0.09
Hierarchical, 25 DDIM	91.8	0.88	0.68	0.26
Hierarchical, 30 DDIM	93.5	1.16	0.90	0.39
Hierarchical, 45 DDIM (ours)	92.1	**1.12**	**0.92**	**0.34**
Hierarchical, 50 DDIM	93.4	1.12	0.83	0.37
(B) Stage 3 elbow. Cols: Z-rng, L-R asym (mm), ER-jerk.				
Functional, GT-train	82.2	0.07	1.8	0.06
Functional, Pred.-train	82.4	0.02	1.4	0.07
Anatomical, GT-train	86.2	0.75	0.5	0.31
Anatomical, Pred.-train (ours)	**82.9**	**0.67**	0.9	**0.25**
Anat., Pred.-train, *MDM*	293.9	6.50	14.1	12.25
(C) Stage 1 wrist loss. Cols: Z-err, ZRngR, ZStdR.				
L1 + velocity	52.8	19.0	0.47	0.50
+ speed matching	52.6	19.1	0.34	0.33
+ **Z log-ratio (ours)**	**62.6**	24.9	**0.85**	**0.73**
(D) Stage 4 lower body. Cols: Pelvis MPJPE, Legs MPJPE.				
Mean-offset baseline	316	310	320	–
FiLM + ARIA	77.7	49.7	96.4	–
FiLM + od	71.1	48.9	85.8	–
BiLSTM + od	73.1	48.4	89.6	–
FiLM spine-only (ours)	**70.7**	**48.5**	**85.6**	–

Pianoid median MPJPE $m_p{\leq}95$ mm, 3 failure with high test MPJPE), spanning Henle grades 3–8 across 12 pieces and both test performers. Six pair types (GT vs. Pianoid, plus Pianoid vs. each of MDM-adapted [17], EDGE-adapted [14], Li2018 [6], and two ablations) yield 72 trials/participant ($\sim$ 30 min) under a naturalness prompt. Analysis uses Bradley–Terry latent ranking [32] and a binomial GLMM (per-participant + per-clip random intercepts) with Holm–Bonferroni correction. A parallel expert leg (N=9 practicing pianists, 10–30 yr exp., median 17.5, recruited independently) probes the GT–Pianoid gap with 6 stratified probes (non-expert 2AFC extremes, MPJPE-best/worst outside non-expert set, anticipatory motion [24, 27], phrase release [26]), with full prompts on the project page. MPJPE-extreme and anticipation/phrase-release probes were blinded to non-expert results, and stratification was fixed before any expert response, ruling out post-hoc cherry-picking. First author coded inductively [33] with advisor review (investigator triangulation), and four themes saturated by the seventh session [34–36].

Twenty-eight non-expert participants (19 with prior piano lessons, 9 amateur-level, none professional pianists) plus 2 partial sessions yielded n=313–315 trials per pair-type. Pianoid beats every non-GT condition. Preference rates are 93.3% vs. MDM, 91.1% vs. BiLSTM, 94.9% vs. Li2018, 89.5% vs. EDGE, and 87.2% vs. single-stage MDM+zs (Holm-adj. GLMM, $p<0.001$). Bradley–Terry places Pianoid (-1.28) between GT (0) and all baselines/ablations (-3.20 to -4.19). GT wins 78.3% head-to-head. Krippendorff α=0.43, in line with reported ranges for subjective perceptual tasks [37]. Per-baseline preference exceeds 0.87 across four GLMM/bootstrap sensitivity variants.

Expert participants (N=9) surface four themes, ordered by agreement. The dominant theme T2 ($7/9$) is lost anticipation before sforzando attacks together with absent vertical wrist articulation on repeated octaves. T3 ($5/9$) is mistimed trunk drift on lyrical probes, a phase rather than amplitude deficit (Pianoid's trunk carries $2.22\times$ GT energy). At lower agreement, T4 ($4/9$) is an attentional-anchor split (non-experts watch trunk/head, experts watch wrist/gaze/elbow), and T1 ($3/9$) is hidden signatures (absent gaze, elbow rigidity). Motion-spectrum analysis corroborates T2 (wrist 8–25 Hz energy ratio 0.03) and T3 (trunk-audio 2 Hz cross-correlation, GT wins $5/6$ at p=0.11, underpowered at n=6).

Δ_1's strongest evidence is the two MPJPE-extreme probes, where joint-position error and expert judgment dissociate. $7/9$ experts reject the MPJPE-best clip (62.7 mm) citing T2, while $5/9$ accept the MPJPE-worst one (180 mm). Δ_1 orders the pair as experts do (0.58 against 0.60) where MPJPE inverts the ranking.

6. DISCUSSION AND CONCLUSION

Pianoid demonstrates two transferable principles for music-conditioned motion synthesis. The first is data-driven body partition. F3's anatomical regrouping inverts the SMPL/MMM "arms" partition that prior hierarchical motion synthesizers inherit, restoring elbow Z-range $10\times$ over the functional partition. The second is selective stochastic generation, with diffusion reserved only for the one-to-many torso sway and deterministic regression elsewhere. The resulting MPJPE trade is principled. MPJPE rewards mean-collapse, which Δ_1 exposes. Non-expert viewers (N=28, 87–95%) and pianist experts (N=9) confirm the trade. Experts localize the residual GT gap to phase mistiming. Δ_1, unlike MPJPE, orders the MPJPE-extreme probe pair as experts do.

Both principles should transfer to domains with similar structure. Violin bowing (one-to-many phrasing under fixed pitch), drumming (constrained targets with expressive elaboration), and conducting (free upper body on score) are natural candidates, and Δ_1 transfers wherever expressive synthesis risks mean-collapse.

Limitations. Evaluation is single-corpus (MOSA, Western classical, 8 pianists), so the two principles are demonstrated rather than shown to generalize, and unseen-piece results rest on $12/80$ samples. Per-note velocity is not annotated, which likely explains the residual sforzando deficit and means the Stage 2 sampler supplies dynamics the conditioning never specifies. We ablate grouping and generator per stage but do not sweep alternative stage orders or partitions, so the split is justified rather than proven optimal.

Data-derived body partition with selective diffusion makes 3D pianist body motion from MIDI a tractable MIR target.

7. ETHICS STATEMENT

Both user studies were conducted with informed consent and approved by the KAIST Institutional Review Board (KAISTIRB-2025-75). Non-expert 2AFC participants (N=28) and expert interview participants (N=9) provided explicit consent at session start and were compensated for their time. Ground-truth motion is drawn from the public MOSA corpus [8] under its release terms. The corpus covers professional musicians who consented to open research use.

Our model operates at the note level without identity conditioning, but the hierarchical architecture could be adapted for unconsented pianist impersonation. We therefore do not pursue identity-conditioned extensions. MOSA covers Western classical repertoire by conservatory-trained professionals. Expressiveness claims should not be assumed to generalize to non-Western, amateur, or improvisatory traditions. Deployment in production educational or assistive contexts should await broader-corpus follow-up.

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